XIAMEN UNIVERSITY MALAYSIA

ONLINE FINAL EXAMINATION

Course Code:	EEE301			
Course Name:	Signals and Systems			
Question Paper Setter:	Dr. Siti Balqis Samdin			
Academic Session:	2020/09	Question Paper:	A ☑	B ☐
Total No. of Pages:	5	Time Allocated:	2 hours	
Additional Materials:	Appendix A			
Apparatus Allowed:	Scientific Calculator			

INSTRUCTIONS TO CANDIDATES

1. Answer all questions.
2. Please follow the requirement of each section and write in all the corresponding answers on the downloadable answer book provided.
3. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1 (20 Marks) CLO1-PLO1:C2

Answer following questions accordingly.

a) Based on your understanding, discuss the use of Fourier series and Fourier transform in linear time invariant analysis and their relation.

[4 Marks]

b) Determine the complex exponential Fourier series of $x(t)$ given as follow

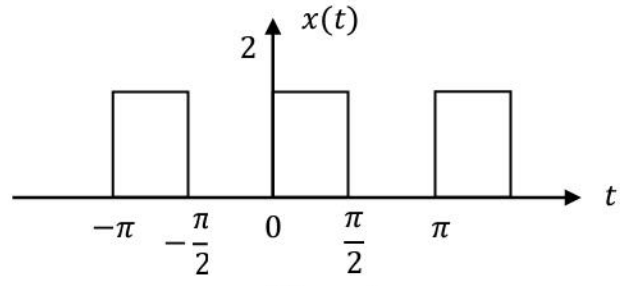


Figure 1

[8 Marks]

c) Determine the Fourier transform of following continuous signals using analysis equation.

i. $x(t) = t\, e^{-2t} u(t)$

ii. $x(t) = e^{-t} \cos t\ u(t)$

[8 Marks]

Question 2 (20 Marks) CLO3-PLO2: C3

An amplitude modulation system in Figure 2.A has an input signal $x(t)$ and carrier signal $m(t) = \cos \omega_0 t$. The spectrum for signal $x(t)$ is shown in Figure 2.B.

a) By using Fourier transformation properties,

i. determine the equation $G(\omega)$ and

ii. sketch the spectrum $G(\omega)$.

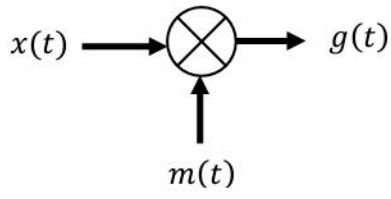


Figure 2. A

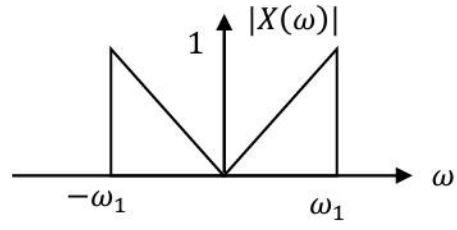


Figure 2. B

[10 Marks]

b) If the modulated signal $G(\omega)$ is being transmitted, suggest a modulation system with filtering in the receiver to get back the input signal form, denoted as $\hat{X}(\omega)$. Explain your system with the aid of diagrams, output spectrums and the frequency domain formulation.

[10 Marks]

Question 3 (20 Marks) CLO3-PLO2: C3

A Butterworth filter has the following characteristics,

i. Band-pass gain between 1 and 0.944 for $0 \leq \omega \leq 100$

ii. Band-stop gain not exceed 0.1 for $\omega \geq 200$.

By evaluating the characteristics, design a transfer function of the filter. Show your calculation step by steps. Then, fill up following Table 1 and sketch your designed filter.

Table 1

ω	0	40	80	ω_p	120	ω_c	160	ω_s	240	280
$\lvert H(j\omega)\rvert$										

[20 Marks]

Question 4 (20 Marks) CLO2-PLO2: C4

The relation of input, $x[n]$ and output, $y[n]$ of a causal discrete linear invariant system is represented in following equation,

$$0.1y[n] - 1.2y[n-1] + 4.7y[n-2] - 6y[n-3] = 2x[n] - x[n-1].$$

By using z-transform approach,

a) Determine the transfer function, $H(z)$ of the system

[3 Marks]

b) Determine and sketch the zeros and poles on z-plane

[6 Marks]

c) Determine the impulse response, $h[n]$.

[6 Marks]

d) Implement the direct form realization of IIR filter with the transfer function obtained in a).

[5 Marks]

Question 5 (20 Marks) CLO2-PLO2:C4

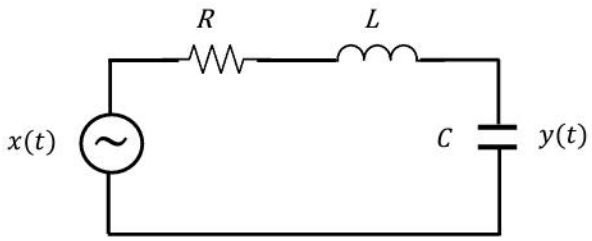


Figure 3

Figure 3, shows a series RLC circuit with input voltage $x(t)$ and output voltage $y(t)$. The input-output relation is described in differential equation

$$\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y(t) = x(t),$$

with the initial condition of $y(0^-) = y'(0^-) = 1$ and the input $x(t) = u(t)$. By using Laplace transform approach,

a) determine the overall output response, $y(t)$.

[10 Marks]

b) determine the transfer function, $H(s)$.

[3 Marks]

c) if the input signal is $x(t) = 5e^{-0.1t}u(t)$, determine the output response $y(t)$.

[7 Marks]

- END OF PAPER –